TVM — Trust Verification Module

Parent Standard: Trust Layer Standard (TLS™)
Category: Governance & Enforcement
Subcategory: Trust Verification
Type: Trust Architecture Module
Version: 1.0
Status: Canonical · Open Module
Effective Date: 10 May 2026
Compatibility: OOF® Methodology OS™ · TLS™ · TVL® · INTEGROS® · UCL™ · OBIDENITY™ · ART™ · ARIV™
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition

Trust Verification Module (TVM) defines the structural conditions under which the trust-state of a system, process, entity, operational environment, or AI-mediated architecture may be verified through traceable evidence, validated conditions, canonical meaning consistency, identity continuity, runtime integrity, and auditable operational behavior.

A system satisfies TVM only if:

- trust claims are testable against structural evidence
- verification does not depend on declaration alone
- trust-state can be linked to identifiable validation conditions
- operational behavior can be compared against required trust conditions
- trust verification remains reconstructable and reviewable

A system that claims trust without verifiable structural evidence does not satisfy TVM.

Module Function

TVM defines the verification layer of trust architecture.

It ensures that trust is not treated as a symbolic statement, institutional assumption, or inherited reputational status, but as something that can be tested against actual system conditions.

The module applies wherever a system claims to be:

- trustworthy
- validated
- reliable
- auditable
- integrity-compatible
- continuously dependable under governed conditions

- TVM™ does not create trust by statement.

It defines how trust must be verified before it may be accepted as structurally real.

Step 1 — Define the Trust Object

The organization must define what exactly is being verified for trust.

Minimum requirement:

- the trust object is explicit
- the object of verification is distinguishable from surrounding systems
- undefined trust targets are excluded from valid verification logic

This may include a:

- system
- process
- module
- entity
- workflow
- AI agent
- operational environment

Step 2 — Define Trust Conditions

The system must define which structural conditions must be satisfied for trust to be considered valid.

Minimum requirement:

- trust conditions are explicit
- required validation conditions are identifiable
- trust does not depend on vague reputation or symbolic assurance alone

Trust conditions may include:

- methodology alignment
- validation status
- integrity enforcement
- identity continuity
- canonical meaning consistency
- runtime traceability
- auditability

Step 3 — Define Verification Evidence

The system must define what evidence is sufficient to
verify trust-state.

Minimum requirement:

- evidence classes are explicit
- verification is linked to traceable sources
- unverifiable trust assertions are excluded from valid trust logic

Evidence may include:

- validation results
- runtime logs
- audit records
- identity continuity proofs
- execution traces
- canonical interpretation consistency
- integrity-state confirmation

Step 4 — Define Verification Method

The system must define how trust claims are tested against evidence.

Minimum requirement:

- verification logic is explicit
- trust-state can be checked against required conditions
- trust verification does not depend on assumption alone

This means the system must be able to answer:

- what is being trusted
- why it is trusted
- on what evidence
- under which conditions
- for how long

Step 5 — Define Verification Boundaries

The system must define the scope and limit of each trust
verification result.

Minimum requirement:

- trust verification is bounded
- verified trust is not falsely extended beyond its evidence base
- partial verification is not misrepresented as total trust-state validity

This prevents overclaiming.

A verified trust-state must remain limited to what was actually verified.

Step 6 — Preserve Verification Traceability

The system must preserve traceability of the trust verification itself.

Minimum requirement:

- trust verification events are reviewable
- verification outcomes remain linked to evidence
- later reconstruction of verification remains possible
- Verification must itself become auditable.
- Otherwise trust verification becomes another opaque claim.

Step 7 — Restrict Invalid Trust Verification

The system must not be treated as valid if trust is declared without reconstructable evidence, bounded conditions, or structural verification logic.

Minimum requirement:

- invalid trust conditions are identifiable
- symbolic trust claims are excluded
- unverifiable trust language is blocked or invalidated where verification is required

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Scenario

An organization deploys an AI-supported operational system and states that it is trusted, validated, and safe for critical decision support.

Application

TVM™ is used to ensure:

- the trust object is clearly defined
- the trust conditions are explicit
- evidence exists for validation, integrity, identity continuity, and runtime behavior
- the trust claim can be tested rather than merely announced

Result

The organization gains a trust architecture in which system trust can be verified against structural reality rather than assumed from brand, certification language, or internal confidence.

Scenario

Multiple entities cooperate across a shared operational environment and must determine whether a participating system or process remains trustworthy under agreed conditions.

Application

TVM™ is used to ensure:

- the trust object is bounded
- the conditions for trust are explicit across parties
- verification evidence remains traceable
- trust is not treated as inherited simply because participation exists

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Result

The environment gains a stronger basis for inter-system trust because trust becomes evidence-based and reviewable rather than symbolic or reputation-dependent.

Canonical Closing Statement

If trust cannot be verified against structural evidence, it remains a claim, not a trustworthy state.

Related Documents

→ Trust Layer Standard (TLS™)

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Canonical Source

https://originopenfoundation.org/